

Tracking Mac Miller's Emotional Evolution Across Albums

Central Argument:

Any Gen Z music fan knows that Mac Miller's lyrical content becomes measurably sadder, more introspective, and less dense over his career. Using word bucket density analysis and statistical testing, I show that his later albums have significantly higher sad keyword density and lower words per minute compared to his earlier work, and that these differences are statistically significant. His music genuinely reflects his emotional evolution over time.

Data Description

What datasets used and why:

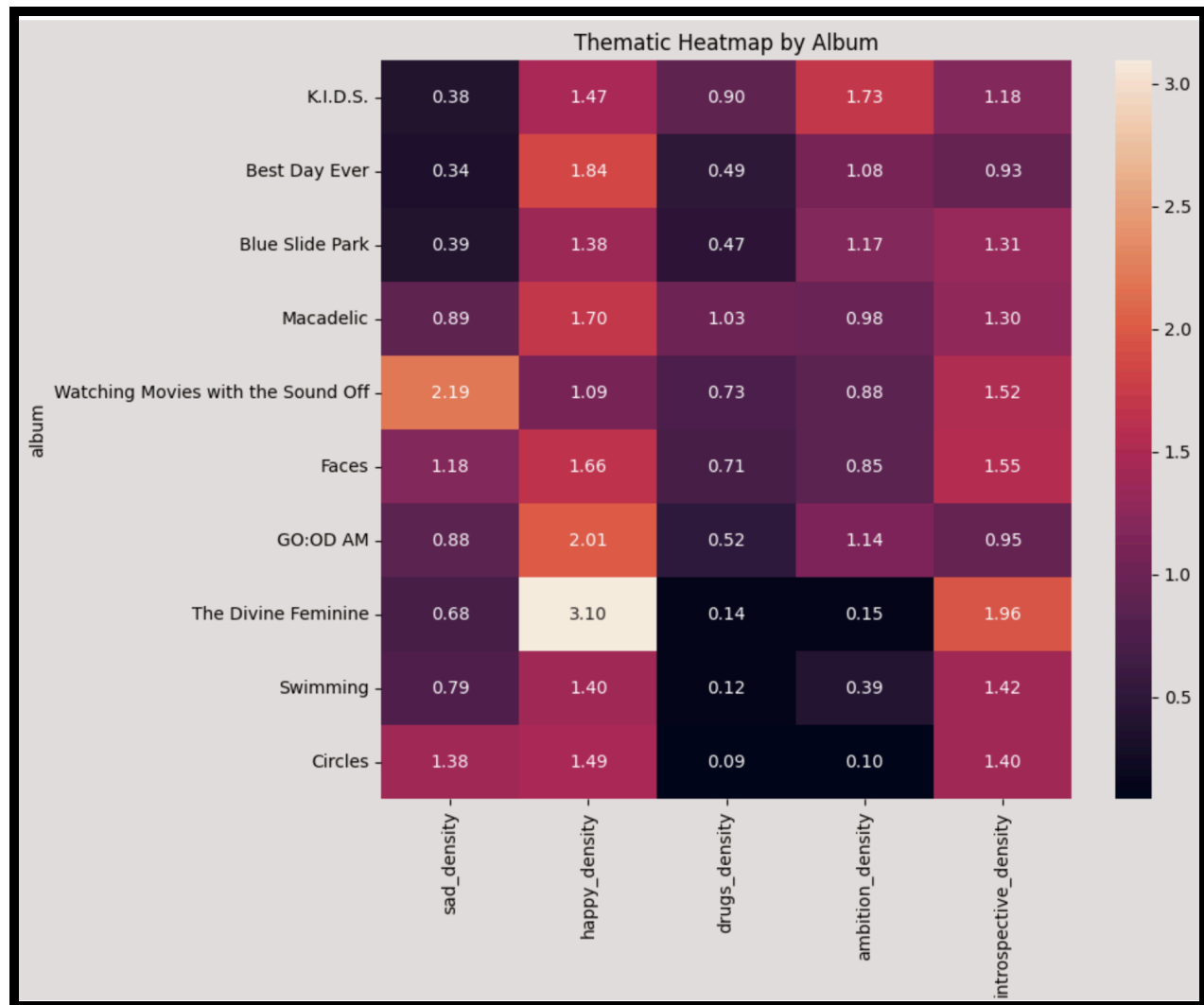
I used a GitHub dataset of 390 Mac Miller songs that includes full lyrics and metadata for each track (album name, release date). The reason I chose this dataset is that my central question is about how his tone changes chronologically across albums, and I needed both the lyrical content and the album/date metadata together. Without the metadata, I can't assign songs to albums or put them in order. I narrowed the 390 songs down to 163 songs from his 10 main studio albums including the posthumous album *Circles*.

How was data retrieved and cleaned:

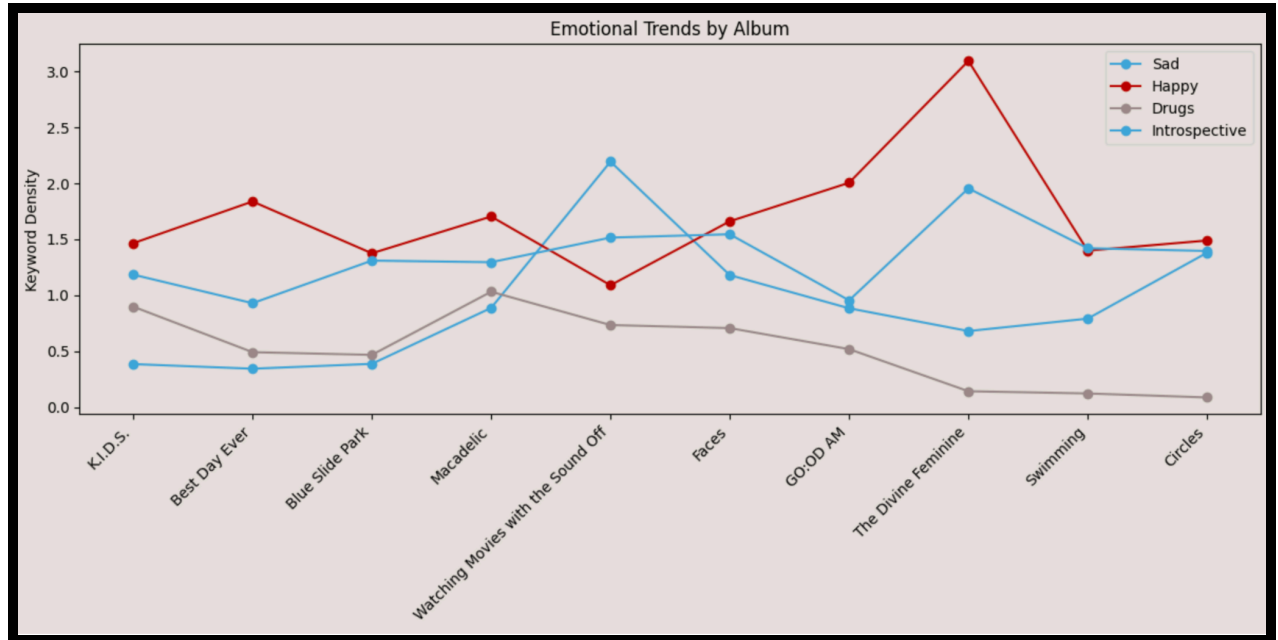
The data was pulled directly from GitHub as a CSV. To clean it, I removed bracketed tags like [Verse 1] and [chorus] that were all over the lyrics. I also stripped special characters, lowercased all text, and calculated word count per song in order to have normalized data. I then normalized by album average duration to get a words per minute metric so longer songs don't inflate the data unfairly. For emotional analysis, I created five keyword buckets: sad, happy, drugs, ambition, and introspective. I built each bucket manually with words that fit the category. For example, the sad bucket includes words like pain, alone, demons, and death. Bucket density is then the percent of total words in a song that match a given category.

Main Methods Used and Outputs Produced

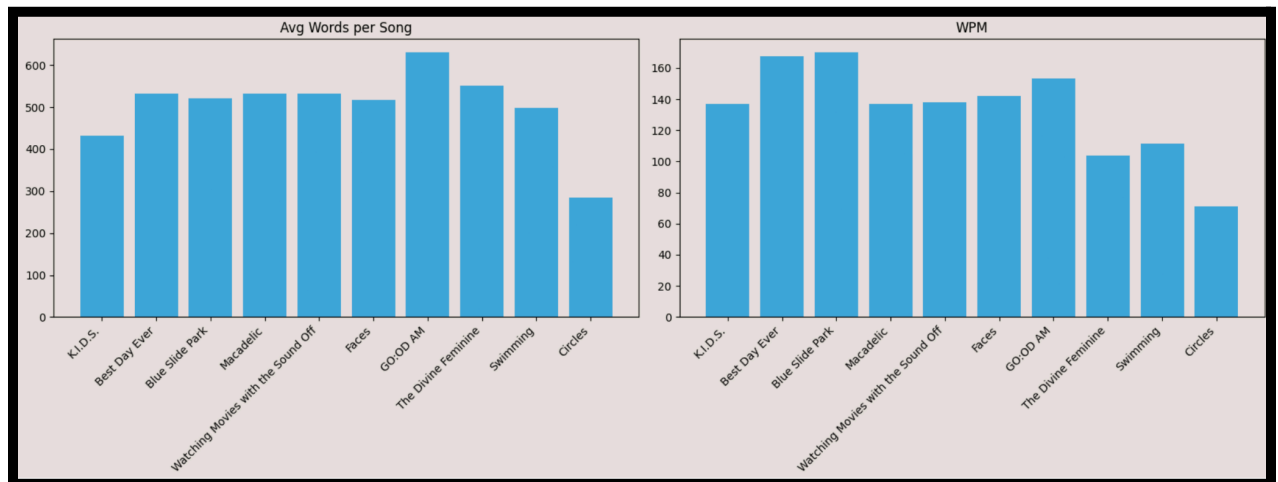
All visualizations are shown before explanations follow.



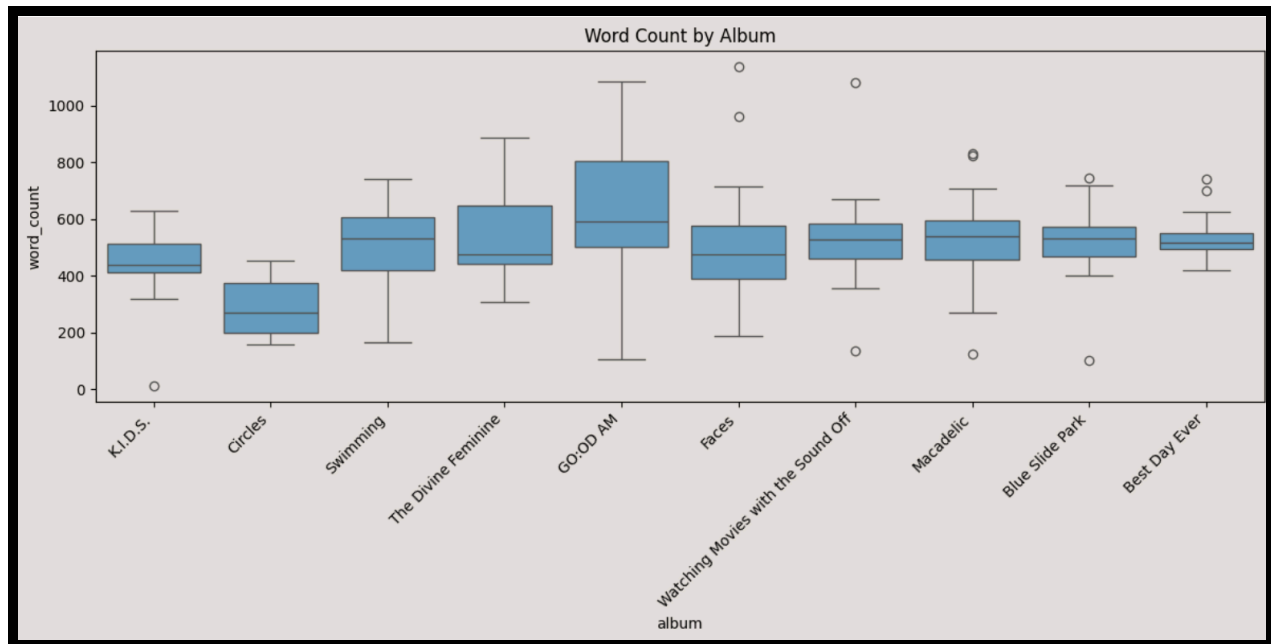
heatmap - keyword density by album



line plot - emotional bucket density over albums



Bar chart - words per song and wpm per album



Histogram - word count per album

Why you chose them:

I chose the line plot because it makes it really easy to see how the emotional categories trend across albums over time. This made it easy to track key buckets such as happy and sad. The heatmap is great for comparing multiple categories at once, since the color makes it obvious which albums score high or low on each bucket without having to read numbers. The bar charts for word count and words per minute let me directly compare lyrical density across albums. I needed to do both because some albums just have longer songs, so the raw word count is misleading without normalization.

What do they show:

The line plot shows that sad and introspective density both rise steadily across his career, while drug and ambition themes decline in later albums. Interestingly, *The Divine Feminine* scores as his happiest album despite coming late in his career. It also scores as the most introspective. The heatmap confirms *Circles* as a unique album with very low drug and ambition scores, signaling a more reflective period. *Watching Movies with the Sound Off* stands out as his lyrically saddest album. The word count charts show that his earlier albums like *GOOD: AM* and *Macadelic* have the highest word count song, and *Circles* has the lowest, which matches the shift in his energy and themes over time. The words per minute normalization was key here. *GOOD:AM* having the highest raw word count makes sense, but once normalized, the gap between early and late albums becomes even more stark. *Circles* sitting at the bottom of both metrics tell the same story

twice. Mac's highest WPM albums were at the start of his career, *Blue Slide Park* and *Best Day Ever*. This shows his ambitious, go-get it attitude that he started his career off with.

Statistical Tests/Regression

Why you chose them:

I used the ANOVA first because I wanted to know broadly whether word count actually differs across albums or if it could just be random variation. From there, I used t-tests to drill into specific comparisons that I actually cared about. The first vs last album comparison is a direct test of the career decline I was seeing visually. The early vs late sad density comparison is the core of my argument, that his music genuinely got sadder and the data confirms it. I looked at 3 of his earliest albums, and 3 of his later albums (*Faces* is technically a mixtape but for the sake of this assignment it is an album). T-tests make sense here because I am comparing two groups and the sample sizes are reasonable.

What do they show:

The ANOVA confirmed there is a statistically significant difference in mean word count across albums (rejected null at $\alpha = 0.05$). The first vs last album t-test confirmed a significant drop in word count from his debut to *Circles*. This confirms my suspicion: Mac's decreased energy was shown in his music, as there became less words from the first to last album. The sad density t-test found that late albums have roughly 3x higher sad keyword density than early albums and this difference is statistically significant. Taken together, these results confirm that the emotional and lyrical trends I observed in the visualizations are real and not just noise in the data.

Limitations of Data and Analysis Approach

The biggest limitation is ambiguity in word classification. A phrase like "sunshine never stays" is sad, but the word sunshine gets counted as happy by my bucket method. The keyword approach misses context entirely. I mentioned this a lot in my presentation, as it is something I have a huge disagreement with. Another issue is that some songs were recorded much earlier than their release date, so the chronological order of albums does not perfectly reflect when the music was actually made. There was also no way to separate guest verses from Mac's own lyrics in the dataset, though he did not feature heavily on most of his tracks. Another limitation worth noting

is that the keyword buckets themselves were built subjectively. Two people could disagree on whether a word like "lost" belongs in sad or introspective, and that call changes the density numbers. The analysis is only as good as the buckets, and there is no perfect way to build them. Finally, this is a lyrics-only study and instrumentation plays a huge role in how his albums feel emotionally. *Swimming* and *Circles* feel like his saddest albums to me (and any fan of Mac's), but the instrumentation is not captured here at all.

Link to Colab:

<https://colab.research.google.com/drive/11kgxoEYJoofw9c-6tDXcQvNavxwLXVsD?usp=sharing>

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